MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
Kokapet(Village), Gandipet, Hyderabad, Telangana – 500075. www.mgit.ac.in
UGC Autonomous | Affiliation to | AICTE approved | Accredited by A++ NAAC | NBA

MOTIVATE INNOVATE EMPOWER 26 YEARS

COMPARATIVE ANALYSIS OF BRAIN AGE AND BRAIN DISEASE

Project Id: IT–19–21

Project Supervisor

Mrs. U. Chaitanya,
Asst. Professor,
MGIT.

Project Member

Shahista Tamkeen
(20261A1249)

OUTLINE

1. Introduction
2. Literature Review
3. Methodology
4. System Requirements
5. Design Phase
6. Coding
7. Testing
8. Deployment
9. Conclusion
10. Q&A

INTRODUCTION

- Welcome to an exciting exploration at the crossroads of neuroscience and artificial intelligence – the initiative titled “COMPARATIVE ANALYSIS OF BRAIN AGE AND BRAIN DISEASE”.
- This endeavor embarks on a quest to forge connections between these two domains, with the overarching goal of unraveling the mysteries surrounding brain aging.
- The human brain, a remarkably intricate and multifaceted organ, undergoes a continuous evolution, mirroring the diverse experiences and interactions that shape an individual's life.
- The concept of brain age, which quantifies how the structure and function of the brain change over time, holds key insights into the natural progression of life, cognitive health, and the understanding of neurological disorders.

OBJECTIVES

- The primary objective of the "COMPARATIVE ANALYSIS OF BRAIN AGE AND BRAIN DISEASE” project is to develop a highly accurate and robust predictive model that can estimate an individual's brain age based on neuro imaging data.
- This predictive model will leverage advanced machine learning techniques, including deep learning and multi-modal data integration, to provide precise and reliable estimations of an individual's brain age.

LITERATURE REVIEW

S.NO	TITLE	YEAR	AUTHOR	DESCRIPTION	LIMITATIONS
1.	Anatomically interpretable deep learning of brain age captures domain-specific cognitive impairment	2022	Chenzhong Yin, Phoebe Imms, Mingxi Cheng.	This study introduces a convolutional neural network (CNN) to estimate BA after training on the MRIs from an independent sample. BA estimation errors are notably lower than those of previous studies.	
2.	Machine learning for brain age prediction: Introduction to methods and clinical applications	2021	Lea Baecker,Rafael Garcia-Dias, Vieira, Cristina Scarpazza,	This paper provide an introduction to the methods and potential clinical applications of brain age prediction.	

LITERATURE REVIEW

S.NO	TITLE	YEAR	AUTHOR	DESCRIPTION	LIMITATIONS
3.	Comparative analysis of machine learning algorithms for multi-syndrome classification of neurodegenerative syndromes	2022	Alberto Merola, Janis Reinelt, Karsten Mueller, Sarah Anderl-Straub.	This paper compared support vector machine, random forest, gradient boosting, and deep feed-forward neural networks for the classification of different neurodegenerative syndromes based on structural magnetic resonance imaging.	
4.	Ten years of brainage as a neuroimaging biomarker of brain aging: what insights have we gained?	2021	I.Beheshti, S. Nugent, O. Potvin, and S. Duchesne	This paper biomarkers of the neuroanatomical agingprocesses exemplifies a new trend in neuroscience in order to provide risk-assessments and predictions for age-associated neurodegenerative diseases at a single-subject level.	

LITERATURE REVIEW

S.NO	TITLE	YEAR	AUTHOR	DESCRIPTION	LIMITATIONS
5.	Classification-Biased Apparent Brain Age for the Prediction of Alzheimer's Disease.	2021	Ali Varzandian, Miguel Angel Sanchez Razo, Michael Richard Sanders	The proposed method is based on a data workflow that include typical data science methods, such as outliers detection, feature selection, linear regression, and logistic regression.	
6.	Increased MRI-based Brain Age in chronic migraine patients.	2022	Rafael Navarro-GonzálezDavid Garcia-AzorinÁngel L. Guerrero-PeralÁlvaro.	The study utilizes the Brain Age framework and machine learning to analyze the impact of migraine on brain aging by predicting age from neuroimaging data.	

LITERATURE REVIEW

S.NO	TITLE	YEAR	AUTHOR	DESCRIPTION	LIMITATIONS
7.	Brain asymmetries from midlife to old Adulthood and hemispheric brain age.	2023	Max Korbmacher, Dennis vanderMeer, Dani Beck, Ann-Marie deLange, Eli Eikefjord.	This paper used a set of magnetic resonance imaging (MRI) metrics derived from structural and diffusion MRI data to evaluate age-related differences in brain asymmetry.	

METHODOLOGY

- Methodological Approach: The strategic approach and processes followed.
- Tools and Technologies Used: Highlight key technologies that played a crucial role.
- Data Collection and Analysis Methods: Summarize the techniques used to gather and analyze data.

METHODOLOGY

- **Data Loading and Preprocessing**:
  - The code loads brain imaging data, including the MRI images and corresponding masks (segmentations).
  - It preprocesses the images by normalizing them to zero mean and unit variance and resampling them to a consistent spacing and size.

- **Model Definition**:
  - The segmentation model is defined using a 3D convolutional neural network (CNN) architecture called SimpleNet3D.
  - This architecture comprises contraction, expansion, and output blocks, following a U-Net-like design.

METHODOLOGY

- **Validation**:
  - Periodic validation is performed during training to monitor the model's performance on unseen data.
  - Validation metrics such as loss are computed to assess the segmentation quality.

- **Testing**:
  - After training, the trained model is evaluated on a separate test set.
  - The model predicts segmentations for test images, and the predicted segmentations are compared against the ground truth segmentations to evaluate segmentation accuracy.

SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS

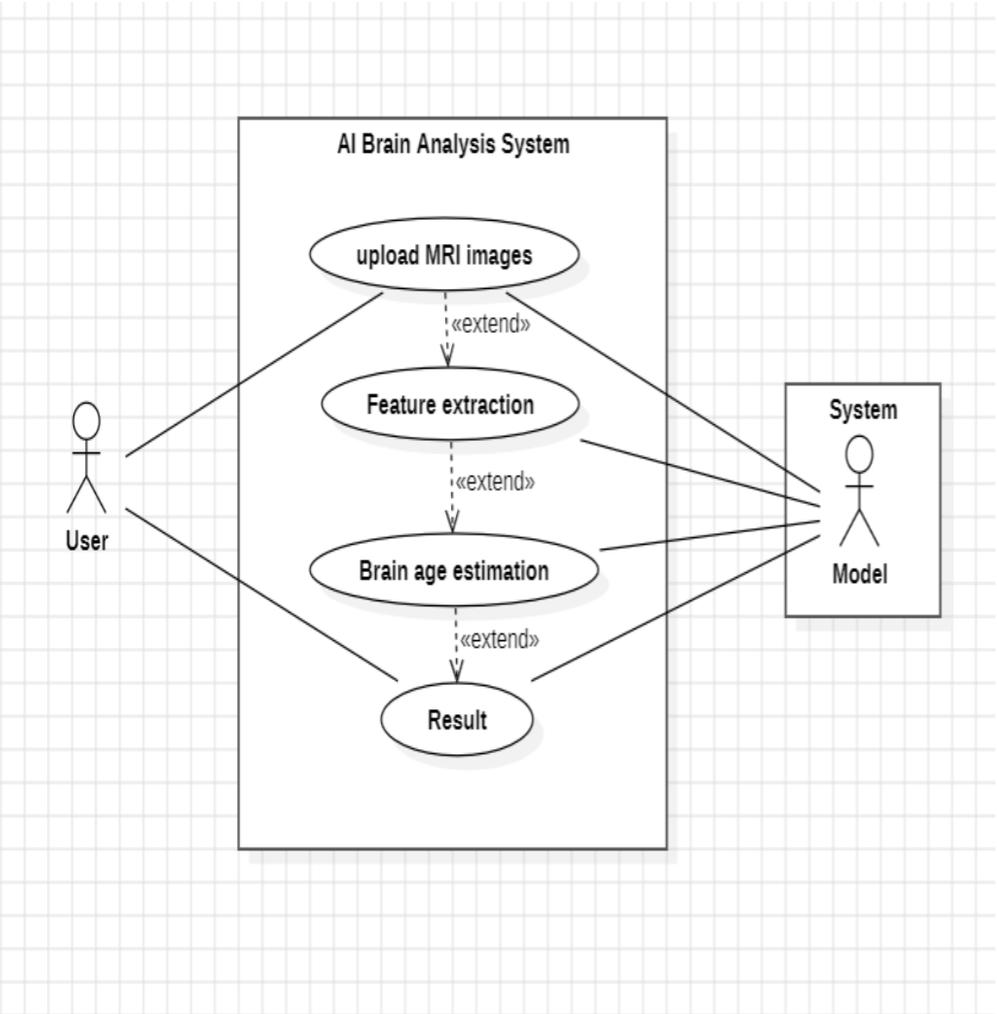
Operating system	: Windows 10
Coding language	: Python
Tools	: Google Collab,Visual Studio Code

HARDWARE REQUIREMENTS

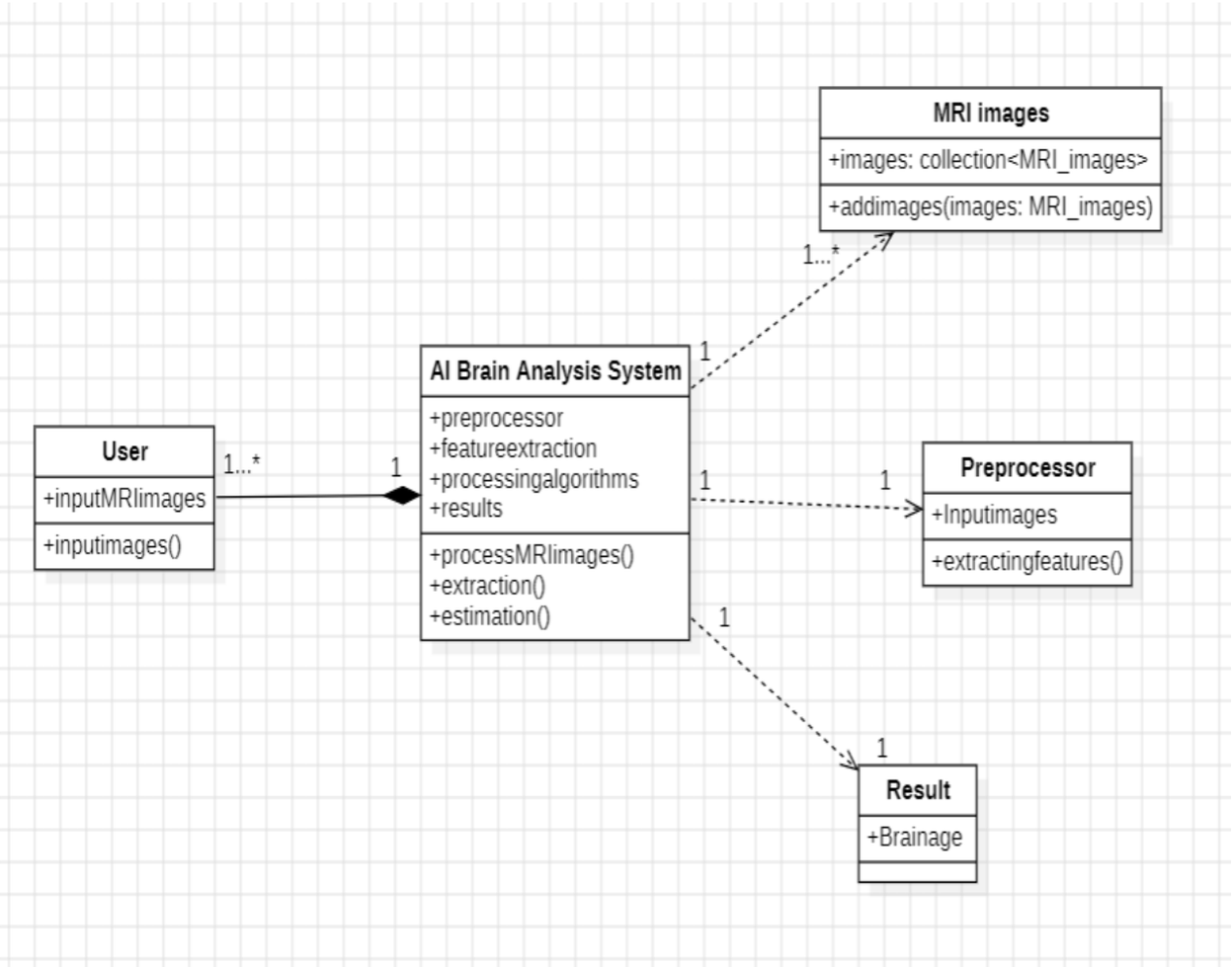
Processor	: Intel Core i7
RAM	: minimum 8 GB

DESIGN PHASE

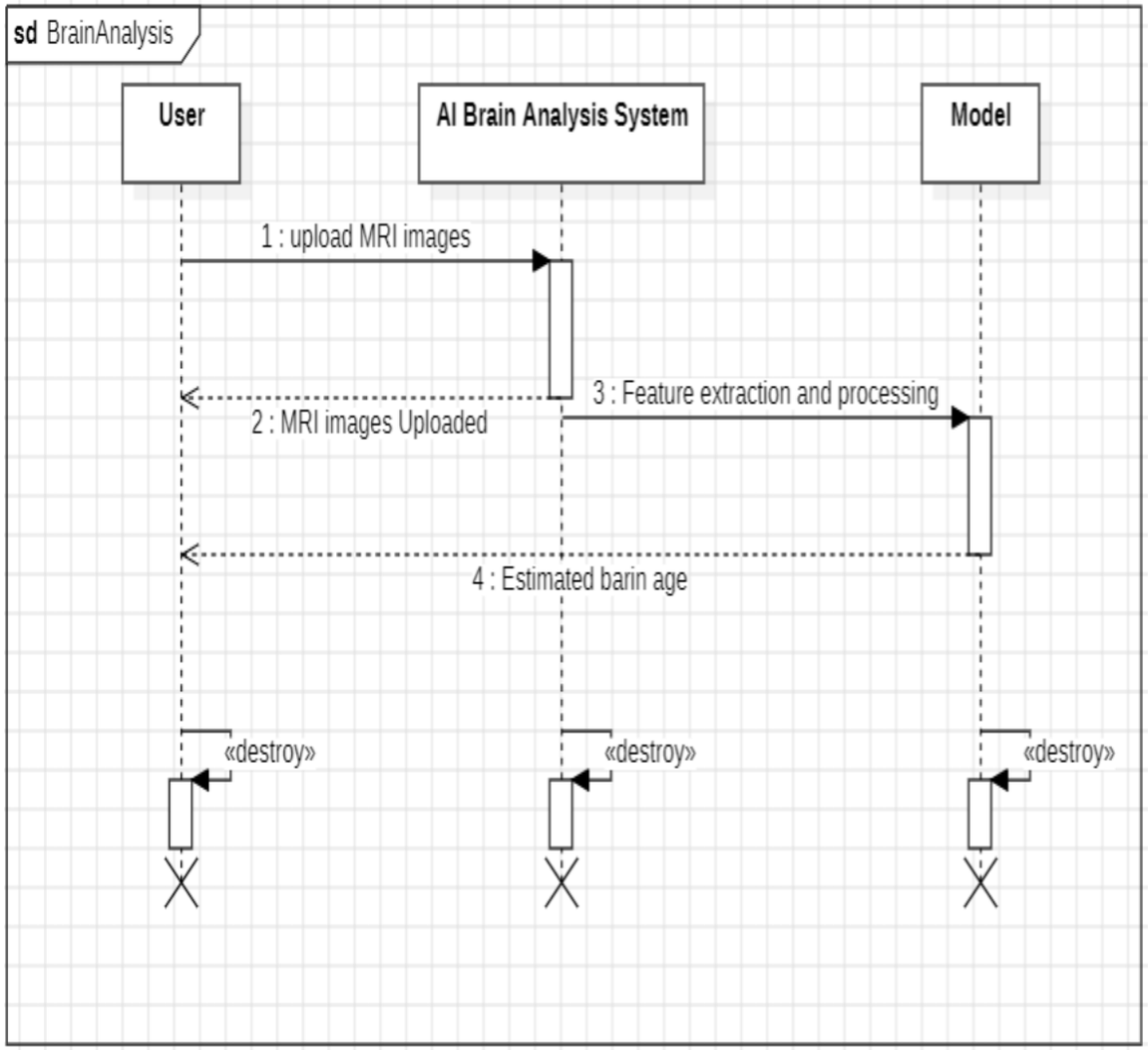
- (One slide for each if complex) Briefly introduce and explain the purpose of each UML diagram:
  - Use Case Diagram
  - Class Diagram
  - Sequence Diagram
  - Activity Diagram
  - State Machine Diagram
  - Component Diagram
  - Deployment Diagram



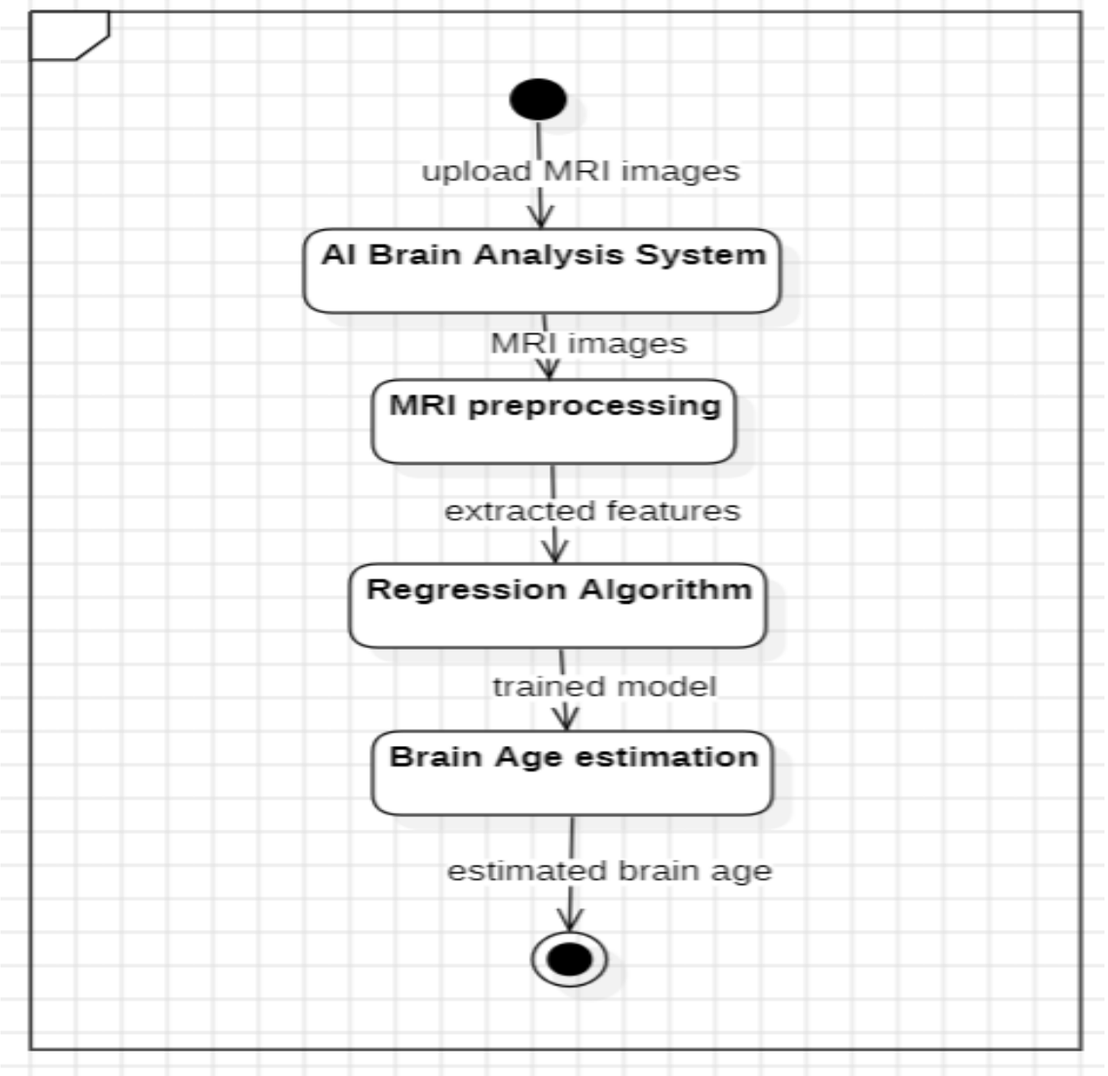
USE CASE DIAGRAM



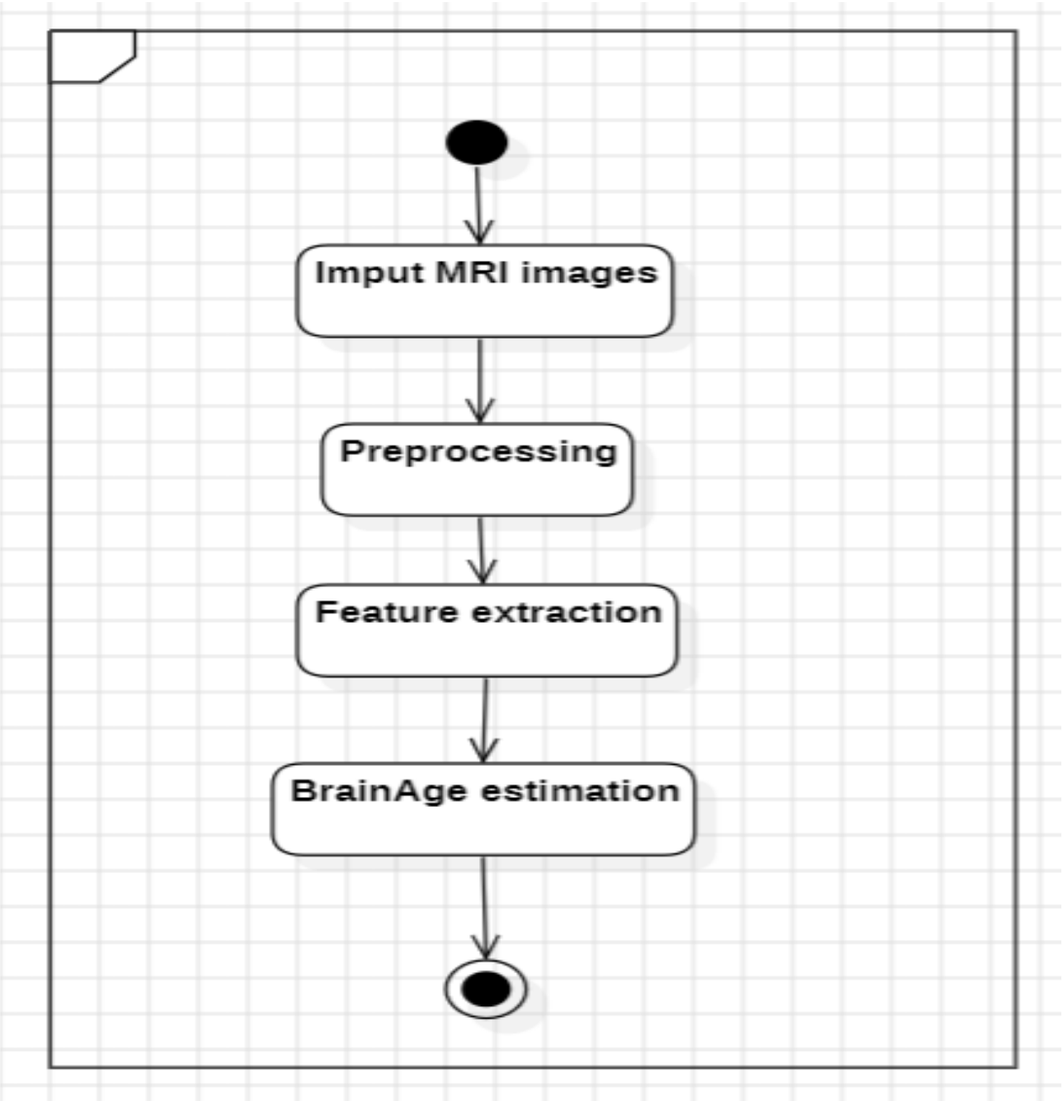
CLASS DIAGRAM



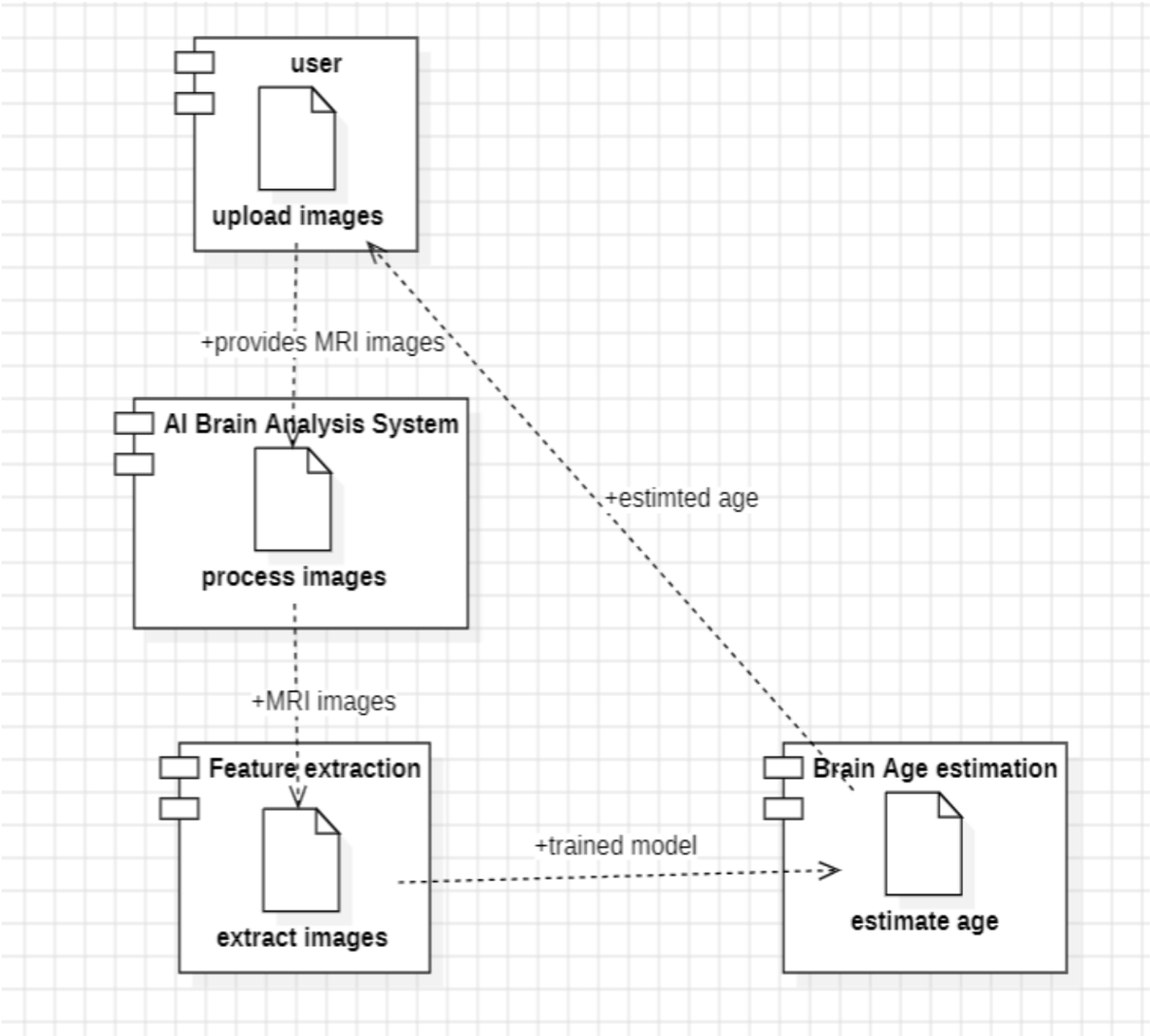
SEQUENCE DIAGRAM



STATE CHART DIAGRAM



ACTIVITY DIAGRAM



COMPONENT DIAGRAM

CODING

```
class SimpleNet3D(nn.Module):
    def contraction_block(self, in_channels, out_channels, kernel_size, padding):
        block = nn.Sequential(nn.Conv3d(in_channels, out_channels, kernel_size, padding=padding),
                              nn.ReLU(),
                              nn.BatchNorm3d(out_channels)
        )
        return block

    def expansion_block(self, in_channels, out_channels, kernel_size, padding):
        block = nn.Sequential(nn.Conv3d(in_channels, out_channels, kernel_size, padding=padding),
                              nn.ReLU(),
                              nn.BatchNorm3d(out_channels)
        )
        return block

    def output_block(self, in_channels, out_channels, kernel_size, padding):
        block = nn.Sequential(nn.Conv3d(in_channels, out_channels, kernel_size, padding=padding),
                              nn.ReLU(),
                              nn.BatchNorm3d(out_channels)
        )
        return block

    def __init__(self, num_classes):
        super(SimpleNet3D, self).__init__()
        self.contract1 = self.contraction_block(1, 64, 3, 1)
        self.maxpool = nn.MaxPool3d(kernel_size=2)
        self.contract2 = self.contraction_block(64, 128, 3, 1)
        self.bottleneck = nn.Sequential(nn.Conv3d(128, 256, 3, padding=1),
                                        nn.ReLU(),
                                        nn.BatchNorm3d(256),
                                        nn.Conv3d(256, 128, 3, padding=1),
```

CODING

```
                                        nn.Conv3d(256, 128, 3, padding=1),
                                        nn.ReLU(),
                                        nn.BatchNorm3d(128)
        )
        self.convtrans1 = nn.ConvTranspose3d(128, 128, 3, 2, 1, 1)
        self.expand1 = self.expansion_block(256, 64, 3, 1)
        self.convtrans2 = nn.ConvTranspose3d(64, 64, 3, 2, 1, 1)
        self.expand2 = self.expansion_block(128, 16, 3, 1)
        self.output = self.output_block(16, num_classes, 3, 1)

    def forward(self, x):
        x_contract1 = self.contract1(x)
        x_pool1 = self.maxpool(x_contract1)
        x_contract2 = self.contract2(x_pool1)
        x_pool2 = self.maxpool(x_contract2)
        x_bottleneck = self.bottleneck(x_pool2)
        x_convtrans1 = self.convtrans1(x_bottleneck)
        x_cat1 = torch.cat((x_contract2, x_convtrans1), 1)
        x_expand1 = self.expand1(x_cat1)
        x_convtrans2 = self.convtrans2(x_expand1)
        x_cat2 = torch.cat((x_contract1, x_convtrans2), 1)
        x_expand2 = self.expand2(x_cat2)
        x_output = self.output(x_expand2)
        return x_output # cross-entropy loss expects raw logits and applies softmax
```

TESTING

- **Functional Testing:** Key results and metrics.
- **Non-Functional Testing:** Results from performance, security, and usability tests.
- **Test Cases:** Example test cases that were particularly significant.

TESTING

Test case:1-Input Image Loading Test Case:

- **Objective:** Verify that the input images are loaded correctly from the specified directory.
- **Expected Outcome:** The image should be loaded without errors and displayed correctly.

Test case:2-Data Preprocessing Test Case:

- **Objective:** Ensure that the data preprocessing steps, such as zero-mean unit variance normalization and image resampling, are performed accurately.
- **Expected Outcome:** The preprocessed image and mask should be correctly normalized and resampled.

Test case:3-Model Training Test Case:

- **Objective:** Validate that the segmentation model is trained properly.
- **Expected Outcome:** The model should converge during training, and the loss should decrease steadily.

TESTING

Test case:4-Model Evaluation Test Case:

- **Objective:** Evaluate the segmentation model's performance on the validation dataset.
- **Expected Outcome:** The model should produce segmentation results with high Dice coefficients, indicating good performance.

Test case:5-Regression Model Training Test Case:

- **Objective:** Verify that the regression model for age prediction is trained effectively.
- **Expected Outcome:** The regression model should achieve reasonable mean absolute error (MAE) and R2 score on the training data.

Test case:6-Regression Model Testing Test Case:

- **Objective:** Test the regression model's ability to predict age on unseen data.
- **Expected Outcome:** The predicted ages should closely match the ground truth ages, with low MAE and high R2 score.

DEPLOYMENT

- Using flask and programming language in python,we have run the web application.

Tools and Technologies:

1.Flask: Flask is a micro web framework for Python used to develop web applications.

CONCLUSION

- The project on the prediction of brain age using machine learning represents a significant stride towards unraveling the mysteries of cognitive aging, fostering personalized healthcare, and advancing our understanding of brain function. Through the convergence of neuroscience and artificial intelligence, this project has aimed to address the complex challenges associated with predicting an individual's brain age and has presented a vision for the future of cognitive health assessment.

Q&A

REFERENCES

1. K. Franke and C. Gaser, "Ten years of brainage as a neuroimaging biomarker of brain aging: What insights have we gained", *Front. Neurol.*, vol. 10, pp. p. 789, 2019.

2. J.H.Cole et al., "Brain age predicts mortality", *Mol. Psychiatry*, vol. 23, no. 5, pp. 1385-1392.

3. I.Beheshti, S. Nugent, O. Potvin and S. Duchesne, "Disappearing metabolic youthfulness in the cognitively impaired female brain", *Neurobiol. Aging*, vol. 101, pp. 224-229, 2021.

4. K. Franke, G. Ziegler, S. Klöppel, C. Gaser and A. D. N. Initiative, "Estimating the age of healthy subjects from T1-weighted MRI scans using kernel methods: Exploring the influence of various parameters", *Neuroimage*, vol. 50, no. 3, pp. 883-892, 2010.

REFERENCES

5. Beheshti, S. Mishra, D. Sone, P. Khanna and H. Matsuda, "T1-weighted MRI-driven brain age estimation in Alzheimer's disease and Parkinson's disease", *Aging Dis.*, vol. 11, no. 3, pp. 618-628, 2020.

6. D. Sone et al., "Neuroimaging-based brain-age prediction in diverse forms of epilepsy: A signature of psychosis and beyond", *Mol. Psychiatry*, vol. 26, pp. 825-834, 2021.

7. Nenadić, M. Dietzek, K. Langbein, H. Sauer and C. Gaser, "Brainage score indicates accelerated brain aging in schizophrenia but not bipolar disorder", *Psychiatry Res.: Neuroimaging*, vol. 266, pp. 86-89, 2017.

8. Beheshti, P. Gravel, O. Potvin, L. Dieumegarde and S. Duchesne, "A novel patch-based procedure for estimating brain age across adulthood", *NeuroImage*, vol. 197, pp. 618-624, 2019.

THANK YOU